

Installation Guide

This guide provides step-by-step instructions for installing and configuring the High-Throughput Point-in-Time (PIT) Feature Store.

Prerequisites

- Python 3.9+
- Docker and Docker Compose
- Apache Kafka 2.8+
- Apache Flink 1.14+
- ClickHouse 21.8+
- Prometheus 2.30+
- Grafana 8.0+

Step 1: Clone the Repository

```
git clone https://github.com/Sarancoding/High-Throughput-Point-in-Time-PIT-Feature-Store
cd High-Throughput-Point-in-Time-PIT-Feature-Store
```

Step 2: Install Dependencies

```
pip install -r requirements.txt
```

Step 3: Configure Environment Variables

Copy the example environment file and update with your credentials:

```
cp harness/.env.example .env
```

Update the following variables in `.env` : - `KAFKA_BROKERS` : Your Kafka broker addresses
- `CLICKHOUSE_HOST` : Your ClickHouse server URL - `FLINK_CHECKPOINT_DIR` : Directory for Flink checkpoints - `PROMETHEUS_ENDPOINT` : Prometheus metrics endpoint - `ENCRYPTION_KEY` : AES-256 encryption key (32 bytes)

Step 4: Start Infrastructure Services

```
docker-compose up -d
```

This starts: - Kafka brokers - Zookeeper - ClickHouse server - Prometheus - Grafana

Step 5: Initialize ClickHouse Schema

```
python storage/clickhouse_schema.py
```

Step 6: Deploy Flink Job

```
flink run -py stream/flink_job.py
```

Step 7: Verify Installation

Run the test suite to verify everything is working:

```
pytest tests/
```

Troubleshooting

Kafka Connection Issues

Ensure Kafka brokers are accessible and the `KAFKA_BROKERS` variable is correct.

ClickHouse Schema Errors

Check that ClickHouse is running and the connection URL is valid.

Flink Job Failures

Review Flink logs at `checkpoints/flink.log` for detailed error messages.